

Solution to Exercise 5.1 (Page 96)

- Illustration of (5.30):

$$ds = \sqrt{dx^2 + dy^2} \quad \text{with} \quad y'(x) = \frac{dy}{dx}$$

$$= \sqrt{1 + y'(x)^2} dx$$

- Objective: $\min \int_0^{s_B} ds \Rightarrow$ Eq. (5.30) gives

$$J(y) = \int_{x_A}^{x_B} \underbrace{\sqrt{1 + y'(x)^2}}_{= \phi(y')} dx, \quad y(x_A) = y_A \quad y(x_B) = y_B$$

- $\phi(y')$ is independent of $y \Rightarrow$ EL eqns. (5.25) simplifies to (5.28)

$$\frac{\partial \phi}{\partial y'} = \frac{y'(x)}{\sqrt{1 + y'(x)^2}} = \text{const.}$$

$\Rightarrow y'(x) = \text{const.} \Rightarrow$ integration yields $y(x) = c_1 x + c_2$

- Determination of c_1 and c_2 with BCs $y(x_A) = y_A, y(x_B) = y_B$:

$$y(x) = \frac{y_B - y_A}{x_B - x_A} x + \frac{y_A x_B - y_B x_A}{x_B - x_A}$$

$\Rightarrow$ minimum distance between A and B is a straight line (as to be expected...)

